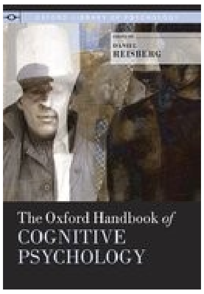

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CHAPTER

39 Induction

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Abstract

This chapter examines recent empirical research and theoretical accounts of inductive reasoning. The first section focuses on property induction between categories. Key phenomena are described and major theoretical models of induction are compared. This section highlights the importance of induction in domains where people have rich background knowledge and research on inductive development in the evaluation of competing models. The next section considers the important but neglected issue of how people make inductive inferences about specific instances. The final sections examine three emerging areas of inductive research: induction when category membership is uncertain, the relationship between inductive and deductive reasoning, and the neural substrates of reasoning. We conclude that future progress in the field will come through the development of broader paradigms that examine induction across a wider range of stimulus domains and models that link induction to other kinds of reasoning and cognitive activities.

Keywords: induction, reasoning, categorization, concepts, knowledge, probability judgment, uncertainty

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Inductive reasoning involves using past observations and knowledge to make predictions about novel cases. Inductive inferences are probabilistic. For example, if you are told that swans, ravens, and swallows all have "sesamoid bones," then you might be reasonably confident, but not absolutely certain, that this property is shared by other birds. Its probabilistic nature distinguishes induction from deductive reasoning, where the goal is to evaluate whether certain conclusions *necessarily* follow from the evidence given (see Chapters 40–41).

Much of the reasoning that people do in everyday life could be described as a form of induction. Predicting the next round of football results, deciding on the most suitable applicant for a job, or inferring whether your children will like a new brand of ice cream, all involve induction. Given its central role in human thought, it is important to understand the cognitive and neural processes that underlie induction. Although the definition of induction is broad enough to encompass a range of psychological phenomena, including object recognition (Nosofsky, 1986), categorization (Anderson, 1991), and word learning (Xu & Tenenbaum, 2007), this chapter will focus primarily on research concerning property induction—how people generalize a novel property from a given category or exemplar known to have this property to other cases. In later sections, however, we will consider the links between property induction and other tasks like categorization and recognition memory (see also Kemp, Shafto, & Tenenbaum, 2012).

There have been quite a few previous reviews of property induction research (e.g., Feeney & Heit, 2007; Hayes, Heit, & Swendsen, 2010; Sloman & Lagnado, 2005). In this chapter we try to avoid going over the
p. 619 same ground by offering a novel way of organizing induction research and theory. In ↳ addition to examining property induction between categories (e.g., generalizing a novel property from dogs to horses, or from dogs to mammals), we will review relevant work on induction between individual instances (e.g., between Rex, our neighbor's dog; and other dogs). This distinction has generally been overlooked in previous reviews. However, we will show that it is critical because these two forms of induction often involve different kinds of perceptual and cognitive processes. Another factor that distinguishes this chapter from previous reviews is our inclusion of developmental as well as adult induction research. Some of the earliest systematic studies of inductive reasoning were carried out with young children (e.g., Carey, 1985). Such developmental data can inform and constrain the development of theoretical models of induction. Our final aim is to familiarize the reader with what we believe to be some of the most important new areas in research on property induction. These include inferences about exemplars whose category membership is uncertain, the relationship between inductive and deductive reasoning, and the neural processes that subserve reasoning.

Property Induction Between Categories

Much of the work on this topic has been carried out using a paradigm introduced by Rips (1975) and refined by Osherson, Smith, Wilkie, and López (1990). The basic method involves presenting a novel property (e.g., "has sesamoid bones") shared by all (or most) members of a category or categories (e.g., "swans" and "ravens"). These categories serve as argument premises. The generalization of this property to other categories is then examined, often by asking people to evaluate the strength of a conclusion (e.g., "that all birds have sesamoid bones") given the premises. In early work the property being generalized was usually unfamiliar or "blank" so that the effect of taxonomic relations on induction could be studied independently of property knowledge. This work has uncovered a number of important principles that influence the relative strength of inductive arguments. Because these have been examined in some detail previously (e.g., Hayes et al., 2010; Heit, 2000), we will not attempt an exhaustive review but instead will concentrate on a number of key phenomena that are particularly important for developing process theories of induction (summarized in Table 39.1).

Key Inductive Phenomena

Premise-Conclusion Similarity and Premise Typicality

The most elementary forms of induction involve generalizing a novel property from a single premise
p. 620 category to a single conclusion. One robust finding ↳ with these kinds of arguments is that the more similar the premise and conclusion categories, the stronger the inductive argument (Osherson et al., 1990; Rips, 1975; Sloman, 1993). This effect has been found across a wide range of category types and seems to be present in some form at a very early point in development (e.g., Welder & Graham, 2001).

Table 39.1 Examples of Some Key Inductive Phenomena Using a Blank Property

Phenomena	Example of a Stronger Argument	Example of a Weaker Argument
Premise-conclusion similarity promotes induction	Orangutans *have property X*/ Gorillas *have property X*	Orangutans *have property X*/ Hippos *have property X*
Premise typicality promotes induction	Crows *have property X*/ Birds *have property X*	Ostriches *have property X*/ Birds *have property X*
Greater diversity and sample size (monotonicity) of positive premises promotes induction	*Diversity* Africans + Americans *have property X*/ All people *have property X* *Monotonicity* Africans + Americans + Europeans + Asians *have property X*/ All people have *property X*	*Diversity* Canadians + Americans *have property X*/ All people *have property X* *Monotonicity* Africans + Americans *have property X*/ All people have *property X*
Causal relations override similarity relations	*Causal violation of similarity* Cheese *has property X*/ Mice *have property X* *Causal nondiversity* Fleas + Butterflies *have property X*/ Sparrows *have property X* *Causal asymmetry* Antelopes *have property X*/ Lions *have property X*	*Causal violation of similarity* Monkeys have *property X*/ Mice *have property X* *Causal nondiversity* Fleas + Dogs *have property X*/ Sparrows *have property X* *Causal asymmetry* Lions *have property X*/ Antelopes *have property X*

Understanding the cognitive mechanisms that drive this effect is not quite as straightforward as it might seem because it presumes that we always know how people evaluate category similarity. There is good evidence, however, that the way people assess the similarity between categories can vary markedly across different judgment contexts (e.g., Medin, Goldstone, & Gentner, 1993), a point to which we will return shortly. Moreover, in developmental research there is an ongoing debate as to whether similarity effects in young children's induction reflect an overlap between the perceptual or conceptual/semantic features of premise and conclusion items (see Gelman & Waxman, 2007; Hayes, McKinnon, & Sweller, 2008; Sloutsky & Fisher, 2004, for contrasting views).

Another important finding is that premise instances that are seen as more typical or representative of a general category lead to stronger inductive projections than less typical premises (Osherson et al., 1990). Hence, a novel property that is true of bears is more likely to be generalized to other mammals than is a property that is true of bats. Again, this effect seems to be quite a general principle in human induction, affecting property inferences in both natural categories and more ad-hoc categories that are defined by a shared goal such as "game" or "friend" (Rein, Goldwater, & Markman, 2010). Like an understanding of the similarity principle, the effects of typicality on induction also emerge relatively early in childhood (López, Gelman, Gutheil, & Smith, 1992; Rhodes, Brickman, & Gelman, 2008).

Notably, little evidence for a corresponding effect of the typicality of the conclusion category has been found (Gelman & O'Reilly, 1988; Rips, 1975; but see Hampton & Cannon, 2004). Gelman and O'Reilly (1988), for example, showed that children and adults were equally likely to generalize a novel property of typical-looking chairs to typical (e.g., a bed) and atypical (e.g., a crib) kinds of furniture. Such findings suggest that when people learn new properties of a typical exemplar, they spontaneously infer these properties to other members of the conclusion category, regardless of exemplar typicality.

Premise Diversity and Monotonicity

Adults' inductive inferences are sensitive to the diversity of premise categories. Properties shared by diverse or dissimilar categories that belong to the same superordinate (e.g., lions and goats) are more likely to be generalized than properties shared by similar categories (e.g., lions and leopards) (Feeney & Heit, 2011; Kim & Keil, 2003). In adults, inductive strength is also positively related to the number of premise categories known to share a property, an effect referred to as "premise monotonicity" (Osherson et al., 1990; for a recent review, see Heit & Rotello, 2012).

Premise diversity and montonicity effects are consistent with normative principles for probabilistic evidence, which hold that conclusions are stronger to the extent that they are based on a broader evidential base and on a greater number of observations (Nagel, 1939).[1] It is interesting, therefore, that an understanding of these principles seems to emerge relatively late in development. Although there are some exceptions (e.g., Heit & Hahn, 2001), most studies have found little evidence that children below the age of 8 years appreciate the value of diverse evidence for property induction (Lawson & Fisher, 2011; López et al., 1992; Rhodes, et al., 2008; Rhodes, Gelman, & Brickman, 2009). Recent work suggests that this may be because adults and children have different expectations about category homogeneity (Rhodes et al., 2008, 2009). Children view categories as more homogeneous than adults, expecting that most category members will share the same essential properties (Diesendruck & HaLevi, 2006; Gelman & Bloom, 2007). This could explain their failure to use premise diversity; if lions, goats, and leopards all belong to the same category (mammals) and young children assume that all category members have the same internal properties, premise diversity will have little effect on their inferences. Supporting this interpretation is the finding that explicitly instructing children about within-category variability promotes use of premise diversity in induction (Rhodes et al., 2009).

Causal Relations Override Taxonomic Relations

The induction phenomena discussed so far are driven by taxonomic relationships between premise and conclusion categories. In many situations however, people use more complex relations between categories (e.g., thematic, script-based, ecological, causal) as a basis for generalizing properties (Medin, Coley, p. 621 Storms, & Hayes, 2003; Rehder, 2009; ↳ Sloman, 2005). Causal relations seem particularly important since they can override taxonomic similarity as a basis for induction. A good illustration comes from work by Rehder (2006). In one study participants first learned about members of an artificial category (e.g., Kehoe ants) made up of a number of discrete features. They were then taught a novel property of Kehoe ants (e.g., "has a venom that gives it a stinging bite") and asked whether this property generalizes to novel instances that had many features in common with the training category (high similarity) or few features in common (low similarity). Critically, on some trials participants were told that a particular feature of Kehoe ants was the cause of the novel property (e.g., the stinging sensation was caused by a high concentration of iron sulfate in the venom). When no causal explanation was given, people generalized the novel property based on similarity. When an explanation was given, people generalized on the presence or absence of the causal feature, regardless of overall similarity. Rehder (2006) also found that the presence of a causal explanation could override taxonomic effects like typicality and diversity.

Causal relations are also important for induction with natural categories. Medin et al. (2003) compared the strength of inductive arguments that used identical categories but varied whether the respective categories were presented as premises or conclusions (e.g., a novel property was generalized from antelopes to lions, or from lions to antelopes). When there was a plausible causal relation linking the premise to the conclusion (e.g., lions eat antelopes) property induction was more likely than in the reversed version, even though category similarity was identical in each case. The presence of a strong causal relation can also override standard taxonomic effects like premise-conclusion similarity, diversity, and monotonicity (see Table 39.1 for examples, but see Heit, Hahn, & Feeney, 2005 for a different account of causal nondiversity).

Children as young as 5 years of age use causal relations in induction (Hayes & Thompson, 2007; Hayes & Rehder, 2012; Opfer & Bulloch, 2007) and by 8 years children see causal relations as a better basis for generalizing properties than global similarity (Hayes & Thompson, 2007). There are some respects, however, in which young children's use of causal relations in inductive reasoning differs from that of older children and adults. For example, young children are precocious in their use of teleological causes (i.e., those based on an object function) in reasoning. Both adults and children use such explanations for artifacts (Lombrozo & Carey, 2006), but children appear less selective in their application of teleology, believing that natural phenomena are subject to the same kinds of explanations. For example, while all age groups agreed that "pens are for writing," 4- to 5-year-olds also claimed that "mountains are for climbing" (Kelemen, 1999).

Sources of Relational Knowledge in Induction

Given that adults and older children use a range of relations in induction, an important goal is to identify factors that lead people to rely on one kind of relation rather than another. At least three such factors have been identified so far. The first is experience with the categories and properties involved in induction (see Chapter 35). Experts with training and experience in biological domains are more likely to rely on causal /ecological relations when generalizing biological properties than are novices, who rely mainly on taxonomic similarity (Coley, 2012; Proffitt, Coley, & Medin, 2000; Shafto & Coley, 2003). The same pattern has been found in comparisons of biological induction by people from cultures with a close contact with the biological world like the Itza Maya of Central America or midwestern Native Americans, with urban undergraduates (Atran, Medin, & Ross, 2005; López, Atran, Coley, Medin, & Smith, 1997). Differences in experience and education may also go some way toward explaining certain developmental changes in induction. For example, children may become more selective in their teleological reasoning as they learn about the variety of causal mechanisms that operate in the natural and technological world. This is suggested by research showing that adults with limited education overgeneralize teleological explanations in much the same way as young children (Casler & Kelemen, 2008).

A second factor that can prompt the use of more complex relations in induction is the nature of the property being generalized. The use of familiar or "nonblank" properties in inductive arguments can have a profound effect on the way that people compare categories in induction. Heit and Rubinstein (1994), for example, found that anatomical properties (e.g., "has an ulnar artery") are more likely to be generalized from sparrows to hawks than from tigers to hawks but that for behavioral properties (e.g., "studies its food before attacking") the pattern reverses. Background knowledge about the property appeared to shift the p. 622 dimensions used in assessing ↳ the similarity of premises and conclusions. In the case of anatomic properties, taxonomic similarity is relevant, but in the case of behavior, predation patterns are relevant (see Shafto, Coley, & Vitkin, 2007, for further examples of the selective generalization of different kinds of properties).

A third factor is the amount of time available to think about a given problem. Shafto, Coley, and Baldwin (2007) showed that when participants had more time to make inductive judgments, they selectively used ecological relations between animals for generalizing disease properties and taxonomic relations for generalizing structural properties like "has a gene." When the same inductive judgments were presented under a short response deadline, however, taxonomic relations were used for both kinds of properties (see Heit & Rotello, 2010, for a related finding, in which speeding up a judgment of deductive validity leads to a focus on taxonomic similarity).

These findings also speak to the important question of whether there is a "default" relation for property induction. A conclusive answer to this question requires more extensive examination of property effects across a range of conceptual domains. Based on existing work, however, it seems safe to suggest that taxonomic similarity is used as a default for generalizing novel properties with biological kinds (Shafto & Coley, 2003; Shafto et al., 2007), whereas information about object function is most frequently accessed in induction with artifacts (Bloom, 1996).

Theoretical Models of Category-Based Induction

One consequence of the recent proliferation of induction research has been the development of a range of competing theoretical models of how people do category-based induction. In this section we will briefly review several of the most prominent models, examining their strengths and limitations and, wherever possible, pointing out where they converge on core principles.

Similarity-Based Models

We use the term "similarity-based" to refer to models that were originally designed to explain induction phenomena that only involve taxonomic relations between premise and conclusion categories. Perhaps the best known is Osherson et al.'s (1990) *Similarity-Coverage* (Sim-Cov) model, which assumes that two processes drive the projection of novel properties between categories. The *similarity* component computes the similarity between the premise and conclusion category with maximal premise-conclusion similarity being used when there is more than one premise. The *coverage* component involves computing the average maximum similarity of the premise categories to samples of the lowest level category, which includes both premises and conclusions. For example, in the argument bears/lions, coverage would involve first generating a superordinate that includes the premise and conclusion categories (e.g., mammals) and then computing the average similarity of the premises to instances from the superordinate.

The similarity component is sufficient to explain phenomena such as premise-conclusion similarity. Other phenomena, however, are driven by both components. Premise typicality, for example, arises because more typical premises (like bears) will have a higher average similarity to the more inclusive conclusion category (i.e., better coverage) than less typical premises (like bats). Diversity effects occur because more diverse premise sets like bears and bats provide better coverage of categories like mammal than do less diverse sets like bears and deer.

One strength of Sim-Cov is that it offers a relatively straightforward account of both adult induction and patterns of inductive development. The coverage component in the model involves more complex computations than the similarity component, particularly for arguments where one has to generate an inclusive conclusion. This suggests a developmental progression that begins with an understanding of arguments based only on the similarity component (e.g., premise-conclusion similarity) followed by arguments involving coverage with a more general conclusion supplied (e.g., bears, bats/mammals), followed by arguments where the inclusive conclusion has to be generated (e.g., bears, bats/lions). This predicted progression is close to what has been observed in many developmental studies (e.g., López et al., 1992).

An important limitation, however, is that the construct of "similarity" is underspecified, even though it is identified as a core component of the Sim-Cov model. Applications of the model (e.g., Osherson et al., 1990) operationalize similarity by obtaining global "similarity" ratings between various categories. But it not always clear what such ratings are measuring. They could reflect an assessment of the semantic relationships between categories, but it is also possible that the category labels activate memories for specific instances whose semantic and perceptual features might be compared.

 Like Similarity-Coverage, Sloman's (1993) *Feature-based induction (FBI)* model emphasises the similarity of premise and conclusion categories as a basis for induction. However, it does so without the retrieval of superordinate categories. Instead, the model assumes that individual categories are encoded in terms of their typical features. Property generalization is proportional to the number of features shared by premises and conclusions, divided by a "magnitude" term, which represents the number of features in the conclusion category. A general prediction of the model is that generalization will increase as a function of similarity (defined in terms of feature overlap) between the premises and conclusions but will be reduced by the presence of "rich" conclusion categories that contain many features.

As well as providing a clear specification of how similarity between categories is computed, FBI makes novel predictions that lie outside the remit of Sim-Cov. For example, it predicts a boundary condition for the effects of premise diversity on induction. According to the model, increasing premise diversity will have little effect on inductive projection when the additional premise has few features in common with the conclusions. For example, even though the argument German shepards, blue whales have property X/Moles have property X involves highly diverse premises, and the premises and conclusions belong to the same superordinate of mammals, the model correctly predicts that participants will judge the argument as inductively weak.

A significant problem for FBI, however, is that it fails to give a complete account of the core phenomenon of premise typicality. Previous work on typicality effects has shown that this result obtains even when the respective similarities between the typical and atypical premises and the specific conclusion category are equated (Rips, 1975). But FBI incorrectly predicts that once premise-conclusion similarity is controlled for there should be no independent effect of typicality. Moreover, the model implies that the typicality of the conclusion category should also affect inductive strength, a prediction that has not generally been supported.

Of course, an even more serious problem for both Similarity-Coverage and FBI is that they fail to provide principled explanations of the effects of causal and property knowledge on induction. In part because it is a more flexible model, FBI fares slightly better in this respect. For example, reversals in patterns of inductive projection driven by different kinds of properties (e.g., Heit & Rubinstein, 1994) could be explained in FBI by assuming that features shared by all premise categories would have the greatest weight in computing overlap with the conclusion category. It is not clear, however, how either Sim-Cov or FBI could explain the primacy of causal relations in induction when these are placed in conflict with similarity. More generally, some have argued that the construct of similarity, which is core to Similarity-Coverage and FBI, is not sufficiently constrained to explain induction (Medin et al., 1989; Sloman & Lagnado, 2005). The category members used in inductive arguments can be compared in an infinite number of ways (e.g., orangutans and gorillas are both apes, hairy, and intelligent, but they are also larger than caterpillars, have fewer than six legs, live on the planet Earth, and so on). Models like Sim-Cov and FBI ignore this complexity, assuming that a reasoner always knows which features of premise and conclusion categories are most relevant for a given prediction, and how features should be weighted and combined in similarity computations. Because of these limitations, recent interest has shifted to the broader and more flexible theoretical frameworks reviewed in the next sections.

Relevance Theory

Relevance theory (Medin et al., 2003) is a broad theoretical framework that aims to account for the effects of causal and property knowledge on induction. A key contribution of relevance theory is that it provides some general principles for identifying the kinds of features that should be considered when evaluating the inductive strength of an argument. The core assumption is that this involves a comparison between the *most distinctive* features of premise and conclusion categories and, in the multiple-premise case, a comparison between different premises. The most distinctive relations activated during this process guide inductive projection. All things being equal, causal relations are generally assumed to be highly distinctive, and therefore more likely to influence property induction, than taxonomic or thematic relations. Nonblank properties can suggest the relevant dimensions for comparing premises and premises, so that the use of different kinds of more familiar properties (e.g., biological vs. behavioral) can prime different sorts of inductive projections.

A positive feature of relevance theory is that it applies the same general principles to explain the touchstone effects involving taxonomic similarity as well as the effects of causal knowledge. For example, a property of
p. 624 kangaroos and koalas is less likely to be ↳ generalized to project to "all mammals" than a property of kangaroos and polar bears, because the less diverse set activates a distinctive property ("native to Australia") that is not shared by most instances of the conclusion. The same mechanism, however, predicts that an even more diverse set like polar bears and penguins will be less inductively potent for the conclusion

category of “animals” (because the premises share a distinctive habitat that is not shared by most other animals).

Although the breadth of relevance theory is appealing, a number of its assumptions need to be refined so that it can generate further testable assumptions. Most notably, the process by which the distinctiveness of the features of premise and conclusions is computed, and how this varies with different reasoning contexts, needs to be specified more clearly. Moreover, a number of the key process assumptions of the model (e.g., that comparisons between the distinctive relations of multiple premises precede comparison between premise-conclusion relations) have yet to be tested. Although much of this work is still to be carried out, there have been some recent positive developments. Most notably, some ideas similar to those in relevance theory have been implemented in a formal model (*SimProb*; Blok, Medin, & Osherson, 2007) and successfully applied to predict probability ratings for inductive arguments with nonblank features.

Bayesian Models

Heit (1998, 2000) suggested that induction is best thought of as a form of Bayesian belief revision. His Bayesian model assumes that when generalizing a novel property (e.g., disease X) between familiar categories (e.g., from sparrows to crows), people will access their prior knowledge about the distribution of familiar properties. They will know, for example, that certain properties are true of all birds, including sparrows and crows, but that other properties are limited just to the premise or the conclusion. The question is which of these distributions the novel property most closely resembles. To solve the problem, the Bayesian model treats the premise or premises in an inductive argument as evidence, which is used to revise beliefs about the prior hypotheses, according to Bayes’s theorem. Once these beliefs have been revised, the plausibility of the conclusion is estimated.

This model predicts most of the key results in Table 39.1. It can be extended to nonblank cases by assuming that different kinds of familiar properties cause the retrieval of different kinds of priors. So when asked whether a biological property of hawks is more likely to generalize to sparrows or tigers, people will retrieve prior knowledge about the anatomical properties across these categories, whereas inductions about behavior will prime retrieval of familiar behavioral properties.

Like other early Bayesian models of induction (e.g., Tenenbaum & Griffiths, 2001), Heit’s model is not precise about how prior probabilities are computed. One possibility is that they are proportional to the number of common and distinctive properties retrieved from memory for premise and conclusion categories. Heit also suggested that prior hypotheses could be influenced by placeholders for essential properties of categories that cannot be listed (cf., Medin & Ortony, 1989). It is not clear whether the model gives priority to causal relations over other properties.

Some of these issues have been addressed in more recent Bayesian accounts such as the structured statistical model of Kemp and Tenenbaum (2009). Like earlier models, this approach assumes that the problem of induction can be approached as one of making a statistical inference about the probability of a conclusion given the observed premises. Notably, the model assumes that the relevant priors used as inputs into Bayesian calculus are based on peoples’ intuitive theories. These theories are instantiated as structural models of the relations between categories and prior beliefs about the distribution of features across categories. Different kinds of structured representations will be retrieved depending on the type of property being generalized. For example, when the induction involves taxonomic properties, the default structure is a hierarchical tree. For spatial or quantitative properties categories are represented according to their distance in dimensional space on known attributes relevant to the target property. When induction involves causal properties, then categories are represented in a directed graph (a Bayes net). Each structure leads to the activation of a different set of priors about the distribution of known features. So in the causal case, knowing that a mouse carries a particular disease may lead to retrieval of knowledge of relevant food-chain relations, activating many features of cats.

This model has been successfully applied to a range of induction data sets, which include many of the key findings in Table 39.1. The relative fit of models based on different structured representations depends on the kind of property being generalized. Bayesian predictions based on the taxonomic model of prior knowledge produce a good fit to p. 625 generalization across a range of animal stimuli when genetic properties were used but a poor fit for disease properties. Predictions based on a causal structural model show the opposite trend, providing a close fit to disease properties that could be propagated via predator-prey relations but a poor fit to inferences about biological properties.

The structured statistical approach provides a clear mechanism for deriving prior probabilities from background knowledge, and it allows for the flexible application of different kinds of knowledge depending on the conceptual domain and nature of the property being generalized. Not surprisingly, there are still some effects of knowledge on induction that the model has difficulty with. Although it can deal with causal relationships that can be represented in a chain or web, it is not clear how the model would deal with cases of induction where premises and conclusion categories are seen as similar because they fulfil similar causal *roles* (e.g., hawks and tigers are similar because they both take the role of predator in their respective environments).

Perhaps the main weakness of the structured statistical model is a by-product of one of its strengths, namely its flexible application of different knowledge structures. The model says little about exactly how people ensure that they activate the correct structured representation for a given problem. When told that mice have a certain disease, it seems likely that people would consider a range of possible routes for property generalization (taxonomic, predator-prey relations, ecological relations, etc.), each of which is associated with a different structured representation for generating priors. Exactly how a particular representation is selected (and other representations discarded) remains to be specified.

Summary and Synthesis

This review suggests that most induction models could be located on a continuum. At one end we have models like Sim-Cov and FBI that are relatively well specified in their processing assumptions and have been formally implemented, but which are relatively circumscribed in scope. These models give a good account of induction in the absence of detailed knowledge about the distinctive properties of premise and conclusion categories. At the other end we have approaches like relevance theory that have a broad scope but whose processing assumptions have yet to be fully specified or tested. An important development in the field, therefore, is the appearance of models like Structured Statistical Representations, which incorporate many of the strengths of the alternate approaches; the model's core assumptions are specified precisely, but it has the scope to explain both elementary and more complex forms of induction.

It seems likely that the next decade will see the emergence of more precisely specified models that attempt to explain both similarity-based and causal induction. For example, Rogers and McClelland (2004) have proposed a connectionist model of property induction that includes generalization of causal properties. As such models proliferate we will need some principled ways of comparing them and ultimately deciding which provides the best account. First, we will need to examine whether, in fact, these models represent qualitatively different kinds of theoretical explanations. McClelland (1998), for example, has suggested that despite their many apparent differences, connectionist and Bayesian models of cognition may share some underlying assumptions and principles. Second, the trade-off between model complexity and success in fitting the data needs to be carefully evaluated. The Structured Statistical approach, for example, explains a far greater range of results than SimCov but does so at the cost of a considerable increase in complexity.

Selection of the most appropriate model will need to be guided by inferential procedures that take account of differences in computational complexity and flexibility (Pitt, Myung, & Zhang, 2002). Third, future formal models will need to account not only for phenomena summarized in Table 39.1 but also for patterns of developmental continuity (e.g., Hayes et al., 2008) and discontinuity (e.g., Rhodes et al., 2008) in induction.

The most important principle for deciding on the best model of induction, however, is not whether the model accounts for known phenomena but whether the model can generate and explain novel (and preferably counterintuitive) patterns of data. In this respect, heuristic approaches like relevance theory have been especially successful. It remains to be seen whether the more recent formal induction models can also generate novel, testable predictions that lead to new discoveries about the way that people do property induction.

Emerging Directions in Induction Research

This section examines a number of new trends in research on human induction. Some of these topics (e.g., induction with exemplars, induction with multiple categories) are of interest because they involve inductive phenomena that are very common in everyday life but have been overlooked ↳ in previous research. Other topics are of interest because they link induction to other cognitive domains like memory and deductive reasoning.

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Induction With Specific Exemplars

Much of the work discussed so far has been based on studies of property generalization between categories. Far less attention, however, has been devoted to how people generalize novel properties between specific exemplars.[2] One reason for the neglect of exemplar-based approaches in induction is that many current theories assume (at least implicitly) a prototype model of representation, where categories are represented by a summary of the most typical category features (see Markman & Ross, 2003 for a discussion of one such approach). This contrasts with the assumptions of many theories of category learning where the similarity between specific exemplars has been recognized as a critical factor in the way people assign objects to categories (e.g., Allen & Brooks, 1991; Nosofsky, 1986).

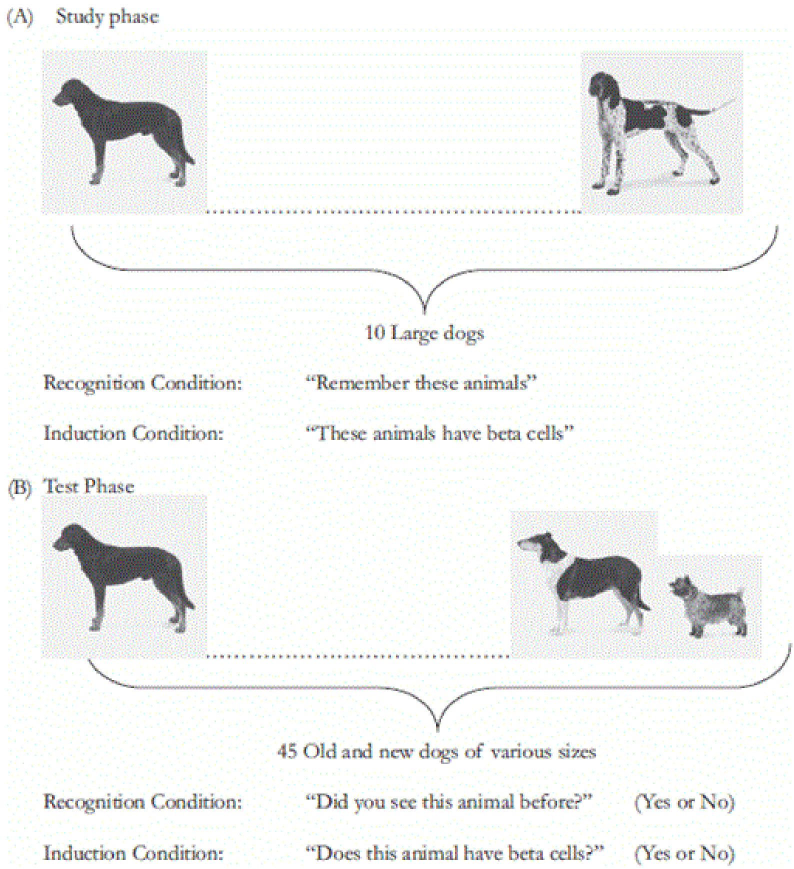
Understanding how people reason about specific exemplars is important for a number of reasons. First, reasoning about specific objects is ubiquitous in everyday life and may involve the consideration of factors that are not usually considered in category-based approaches. If asked to feed our neighbor's dog, Rex, our knowledge of the general properties of *dogs* can only take us so far when making predictions about Rex's behavior and preferences. To make these predictions, we may have to look at Rex's particular features. If Rex is large and barks a lot, we may infer that he is also aggressive toward strangers and should to be approached with caution. Alternately, if Rex is small, beautifully groomed, and has been entered in dog shows, we might predict that he will be a fussy eater. Note that in these cases property predictions are not just based on category-level knowledge. Instead, the predictions are based on knowledge about relationships between features rather than categories (e.g., large, barking dogs are often aggressive).

Is there any evidence that people reason this way? A recent study by Murphy and Ross (2010a) suggests they do. In one study participants were shown artificial categories containing cars that varied in color and in the nature of a "free" feature that came with each particular car (e.g., satellite radio, alloy wheels, heated seat). The categories represented cars produced by different manufacturers (e.g., Honda vs. Ford). The stimuli were designed so that in the crucial condition, one free feature was characteristic of each category overall (e.g., Fords most often had satellite radio). Within the category, however, a different feature was strongly correlated with specific colors (e.g., yellow cars most often had alloy wheels). After hearing about these categories, people were presented with a novel car and told its color (e.g., "I have a new car that is yellow"). They were then asked to classify the instance and to predict what other feature it would have. Even though almost everyone assigned the test instance to the correct category, they did not use category-level information to make feature predictions. Instead, most people used the correlations between specific features as a basis for property induction. This result shows that when people have knowledge about some of the features of an object, they will often use feature-feature relations rather than feature-category relations as a basis for inductive prediction.

Another important reason for examining how people reason about specific instances is that it offers the prospect of linking the study of induction to other important areas of cognitive science. Much of the work in other key domains like memory, attention, and identification has focused on how people make judgments about individual objects rather than categories of objects. This has led to the development of sophisticated models that can explain a wide range of empirical phenomena (e.g., Malmberg, 2008). If the experimental paradigms used to study induction included inferences about individual objects, then we will be in a better position to examine the similarities and differences between the processes involved in induction and those involved in other tasks like object recognition.

To this end we have developed an experimental paradigm that makes induction and recognition memory tasks as comparable as possible (Hayes, Fritz, & Heit, 2012; Heit & Hayes, 2011). The basic approach is illustrated in Figure 39.1. People were presented with a common study set (e.g., pictures of several large dogs) and told that their task was either to learn about animals that shared a novel anatomical property (induction) or to memorize each item (recognition). All participants were subsequently presented with a test set that included study dogs and new dogs. Those in the induction condition were asked to respond "yes" if they thought a test animal had the target property, while those in the recognition condition responded "yes" if they thought they had seen the animal before. In general, those doing induction were more likely to respond "yes" to new test items than those doing recognition. However, the similarities
p. 627 between response ↳ patterns across the two tasks were equally striking. The correlation between the probability of making a "yes" response to the various test items under induction and recognition instructions was very high (0.84 or greater). Moreover, various task manipulations, such as increasing exposure to study items, increasing the length of the study list, and changing the perceptual context in which study and test items were seen, had similar effects on induction and recognition responses. Notably, a close relation between patterns of induction and recognition has also been found in 5–6-year-old children (Hayes et al., 2012).

Figure 39.1



Outline of the procedure used by Heit and Hayes (2011) to compare inductive reasoning and memory. (See color insert.)

These results suggest that induction and recognition share some component processes. One process likely to be shared is an assessment of the specific similarity between test items for which a judgment is required and previously experienced category exemplars. The notion that the total similarity between a test item and previously experienced instances affects judgments about new cases has long been a central feature of formal models of recognition (e.g., Dennis & Humphreys, 2001) and categorization (Nosofsky, 1986). The value of incorporating this principle into models of induction is illustrated by the fact that a modification of Nosofsky's (1986) Generalized Context Model, which relies primarily on an assessment of the total similarity between old and new exemplars, produced an excellent fit to both the induction and recognition data in Heit and Hayes (2011).

There are two important implications of this work. First, it provides clear evidence for the overlap in underlying processes between induction and other well-studied cognitive phenomena. Second, it suggests that in order to deal with induction based on specific exemplars, future models of induction will have to build in mechanisms for comparing the similarity between instances as well as categories.

Induction With Uncertain Categorization

Much of the research on induction reviewed so far has been concerned with making inferences about objects whose category membership is known with certainty. If you know an object is an apple, then you can be p. 628 reasonably certain that it will taste sweet ↳ and have small seeds. If a persistent cough is known to be caused by bronchitis, one can predict that it will eventually subside. In many cases, however, we may have to make inductive predictions about instances whose category membership is ambiguous. Even though bronchitis may be a likely cause for the cough, other causes, such as early-stage lung cancer, are possible alternatives. Clearly our predictions about these cases (e.g., future symptoms and best course of treatment) will depend on whether we consider only the most likely category assignment or whether we factor in the properties of less likely category alternatives.

Bayesian theories like Anderson's (1991) Rational model predict that people will consider multiple categories when making feature predictions about objects whose category membership is uncertain. However, a considerable body of research with natural (Murphy & Ross, 1994, 2010b) and artificial categories (Ross & Murphy, 1996) suggests that people rarely do this; instead they usually ignore the uncertainty and make predictions based on the properties of the category to which an object is *most likely* to belong (referred to as the "primary category").

To illustrate, consider the artificial disease categories of Burlosis and Terragitis in Table 39.2. Each category contains eight "patients" with two characteristic symptoms. Imagine that a new test instance (e.g., a patient with a *headache*) is encountered and you are asked to predict another symptom that the patient is likely to have. Burlosis is the primary category because it contains more patients suffering from headaches. If only this category were considered, then the most likely symptom would be "rash" (as this is the most common symptom on dimension *d*2 within Burlosis). Note, however, that some patients with Terragitis also have headaches. Hence, according to the Rational model, the properties of this secondary category should also be considered, leading to the prediction of "sore gums" (the most common symptom on *d*2 across both categories). When people are presented with such problems, they readily acknowledge that the uncertainty regarding the category membership of the test instance. Nevertheless, most participants ignore this uncertainty, basing their predictions only on the features of the primary category (e.g., Hayes & Newell, 2009; Murphy & Ross, 1994, 2010b).

Exactly why people violate Bayesian prescriptions and make "single category" predictions is not yet clear. One possibility is that this reflects a general constraint on hypothetical reasoning, referred to by Evans (2007) as *the singularity principle*. According to this principle, when people reason hypothetically, they prefer to consider only one possible situation at a time. Hence, people will acknowledge uncertainty when deciding on category membership but focus only on the most likely scenario (i.e., the primary category) category when making a feature prediction.

Table 39.2 Examples of Stimuli Used in Studies of Induction With Uncertain Categories

Patient	Symptom Dimension 1 (d1)	Symptom Dimension 2 (d2)	Patient	Symptom Dimension 1 (d1)	Symptom Dimension 2 (d2)
B1	Headache	Rash	T1	Fever	Sore gums
B2	Headache	Sore gums	T2	Headache	Dizziness
B3	Headache	Rash	T3	Fever	Sore gums
B4	Headache	Rash	T4	Headache	Sore gums
B5	Headache	Sore gums	T5	Fever	Sore gums
B6	Headache	Sore gums	T6	Headache	Rash
B7	Stomach pains	Rash	T7	Headache	Dizziness
B8	Headache	Runny nose	T8	Fever	Sore gums

Induction test: "Patient X is a new patient. This patient has a headache. What other symptom are they most likely to have?"

Multiple category prediction = "sore gums"; *Single category prediction* = "rash."

Source: Adapted from Hayes & Newell, 2009.

p. 629 As with other cognitive heuristics (e.g., Kahneman & Tversky, 1973), however, there is evidence that people can overcome their tendency to make single-category predictions when the other category alternatives are seen as especially relevant or important for the prediction. Hayes and Newell (2009), for example, presented inductive problems like the one in Table 39.2 but told some participants that the less likely category (Terragitis in this example) was "serious and potentially fatal." This led to a marked increase in predictions based on information from both categories. Ross and Murphy (1996) found that when a less likely category alternative was seen as particularly relevant to the feature being predicted, then that category was considered when making feature predictions (e.g., when the feature was whether a stranger walking up a driveway would "pay attention to the sturdiness of the doors" and the category alternative was "burglar"). Hayes and Chen (2008) have also found that certain kinds of domain expertise can promote use of multiple categories for inductive predictions relevant to that domain.

Another important finding is that when people are faced with making inductive predictions about instances whose category membership is uncertain, they may abandon the use of categories altogether. Papadopoulos, Hayes, and Newell (2011) presented artificial categories with a statistical structure similar to the example in Table 39.2 except that across both categories certain features were highly correlated (e.g., headache was frequently paired with the symptom "difficulty swallowing"). The categories were designed so that feature predictions based on these conjunctions were distinguishable from those based on either single or multiple categories. When presented with a test patient with a headache, people overwhelmingly made inductive predictions based on feature conjunctions rather than using either of the category-based strategies (see Griffiths, Hayes, & Newell, 2012; Murphy & Ross, 2010a for similar findings). This again highlights that when making feature predictions about individual objects, people often rely on similarity to specific instances rather than category-level information.

It seems likely that we will see more research on inductive inference with uncertain categories in the coming years. One reason is that this work has the potential to strengthen the links between research on inductive reasoning and areas such as decision making and judgment under uncertainty (see Chapters 38 and 43). A second reason is that this work has some important implications for theories of induction. Induction with uncertain categories appears to involve factors that lie outside the scope of the Bayesian theories (e.g., Kemp & Tenenbaum, 2009) that have been successful in explaining other forms of category-based induction. Finally, this work is important is because it is closely linked to an even more fundamental problem in induction; how people make inferences about cross-classified objects. All objects are members of multiple categories. For example, *Hilary Clinton* is secretary of state, the wife of a former president, a Democrat, an American, and a woman. An important question that remains unresolved is how people coordinate information from these different categories when making feature predictions (but see Hayes, Kurniawan, & Newell, 2011; Murphy & Ross, 1999 for some possible answers).

Relations Between Induction and Deduction

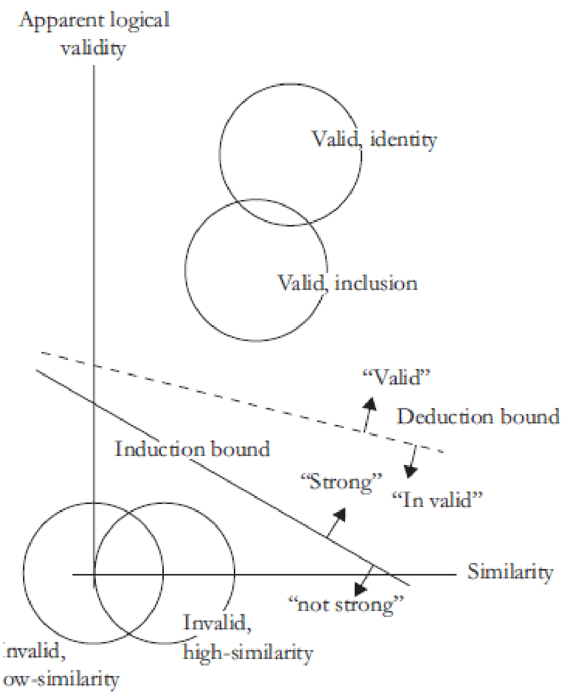
Inductive reasoning is not the only kind of reasoning. The traditional alternative to induction is deduction, which is linked to making valid inferences that are 100% certain given a set of premises and do not depend on background knowledge. Deduction appears very different from induction, which is probabilistic and knowledge rich. But where do you draw the line between induction and deduction, and how are they related? Heit (2007) distinguished between two approaches, the problem view and the process view. According to the problem view, induction and deduction refer to different kinds of reasoning problems. For example, deductive problems could be defined as those arguments that are valid according to the rules of a well-specified logic, and other arguments could be referred to as inductive problems. Alternately, deductive problems could be defined as those with 100% (or perhaps 99.9%) likely conclusions, inductively strong problems could be those with highly probable conclusions (perhaps 75%–99.9%), and other problems could be considered weak.

Unlike the problem view, the process view is concerned with psychological processes rather than the form of the problem. The question of interest is what processes underlie induction and deduction, and whether these are the same or different. Some researchers have suggested that induction and deduction depend on the same cognitive processes. This approach will be referred to as the one-process view. Several influential research programs embody the one-process view, by applying a common framework to both inductive and p. 630 deductive problems. For example, Osherson et al. (1990) and Sloman ↳ (1993) noted that their models of inductive reasoning could, without additional assumptions, account for some deductive reasoning phenomena. Similarly, Johnson-Laird (1994) explained how mental models theory, typically applied to problems of deduction, can also be applied to problems of induction.

In contrast, according to two-process accounts (Evans, 2007; Stanovich, 2009), both heuristic and analytic processes contribute to reasoning. Both induction and deduction could be influenced by these two processes, but in different ways. Induction judgments would be particularly influenced by quick heuristic processes that tap into associative information about similarity and background knowledge. In contrast, deduction judgments would be more heavily influenced by slower analytic processes that encompass more deliberative, and typically more accurate, reasoning. Two-process accounts have provided an explanatory framework for many results, but in general they have not been implemented computationally.

A recent line of work (Heit & Rotello, 2010; Heit, Rotello, & Hayes, 2012; Rotello & Heit, 2009) has directly pitted one- and two-process accounts against each other, by implementing the accounts as computational models and applying them to experimental data. The method, adapted from Rips (2001), is to take a set of arguments and have one group of subjects judge whether the arguments are deductively valid and another group judge whether they are inductively strong. If different variables affect deduction versus induction, then this can be seen as evidence for the two-process account. Indeed, Heit and Rotello found that logical validity itself has a greater effect on deduction judgments, whereas more superficial variables such as similarity and the length of an argument (the similarity and numerosity phenomena in Table 39.1) have a greater effect on induction judgments. Furthermore, forcing people to make very fast deduction judgments led these judgments to resemble induction judgments, suggesting that analytic processing had been impaired and heuristic processing had been emphasized. Heit and Rotello implemented a two-dimensional model of reasoning (see Fig. 39.2) that successfully accounted for these results by assuming that deduction and induction rely on differing proportions of two kinds of underlying information, which could be the outputs of analytic and heuristic processes. In comparison, a one-process model, simply assuming a single underlying process for evaluating argument strength, could not account for the results. It appears that by varying instructions as

Figure 39.2



Two-dimensional model of reasoning, showing arguments varying in apparent deductive correctness (y-axis) and similarity information (x-axis). The dotted line shows the decision boundary for judging whether an argument is deductively valid or invalid. The dashed line shows the decision boundary for judging whether an argument is inductively strong or weak. (Adapted from Heit & Rotello, 2010.)

well as speed of responding (see also Shafto et al., 2007), there will be a rich set of results that will be an important test bed for developing and testing models of reasoning.

Neuroscience of Induction

Technological advances such as functional magnetic resonance imaging (fMRI) have spurred research on the neuroscience of reasoning, with the potential to address important theoretical questions in reasoning research. Researchers use fMRI to measure blood flow related to neural activity in the brain—sampling every few seconds from different positions and producing detailed, three-dimensional images of brain structure and activation (Henson, 2006). Although it seems plausible that induction takes place throughout the brain, given the probabilistic and context-dependent nature of induction, some research has aimed to localize some of the processes involved in induction (and deduction) in regard to particular brain regions. Henson (2006, p. 64) referred to this method as "forward inference," namely "the use of qualitatively different patterns of activity over the brain to distinguish between competing cognitive theories."

p. 631 Several studies (Goel & Dolan, 2004; Goel, Gold, Kapur, & Houle, 1997; Osherson et al., 1998; Parsons & Osherson, 2001) have compared deductive and inductive reasoning tasks, in an attempt to compare one- and two-process accounts of reasoning. To the extent that qualitatively different patterns of brain activity are observed for deduction versus induction tasks, holding everything else equal between experimental conditions, by forward inference, two-process accounts will be supported over single-process alternatives. Indeed, these four studies found somewhat different patterns of brain activation for deduction versus induction. For example, three (Goel & Dolan, 2004; Goel et al., 1997; Osherson et al., 1998) found increased activation for induction, relative to deduction, in left frontal cortex, although in different regions at a finer level. Overall, these results help make the case for two-process accounts over one-process accounts, although there are still unanswered questions, such as why different brain regions are activated in different induction tasks. (See also Goel, 2009). More generally, as Henson (2006) notes, forward inferences are theory dependent: These are used to contrast the predictions of an opposing pair of theories, but it is always possible that neither theory is correct and that some third theory is the best explanation of the results (see Hayes et al., 2010, for further discussion of neuroscience studies of induction).

Concluding Comments

Having reviewed some of the highlights from four decades of experimental and developmental research on induction, increasingly sophisticated models, and the newest work on the neuroscience of reasoning, we find ourselves enthusiastic about the progress that has been made. At the same time, we are aware of the limitations of induction research so far. Along with progress on incorporating rich domain knowledge into reasoning, we believe that models of induction need to become broader and more widely applied. One way that research on induction ought to become more ambitious is in terms of the induction task itself. Many of the experiments we have reviewed involved learning that a set of animals has a hidden property and inferring whether other animals have that property. Yet induction is much broader than that. For example, Heit and Nicholson (2010) have suggested that political reasoning is formally equivalent to some already-studied inductive tasks, and that models of induction might also be applied to reasoning by elected officials in terms of which legislation they will support, or decisions by voters on which candidate to support. Thinking even more broadly, because induction is closely connected to other forms of reasoning, such as deduction (Heit & Rotello, 2010; Heit et al., 2012; Rotello & Heit, 2009), and other cognitive activities, such as categorization, memory, and decision making (Hayes & Newell, 2009; Heit & Hayes, 2005), we pose the question of whether in the future it will be sufficient to focus on induction alone. We suggest that the next generation of theoretical models should be more ambitious, spanning multiple reasoning tasks, such as induction and deduction together, or even multiple cognitive activities, such as induction, categorization, and memory.

Notes

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Notes

1. Some readers may be struck by the contrast between the relatively normative way that adults use sample size and information diversity in inductive reasoning and the range of biases that have been revealed in studies of adult probabilistic judgments (see Chapter 38 for a review). One reason for this discrepancy may be that, in contrast to the tasks often used in studies of judgment, the probability information in most induction tasks is relatively straightforward. Novel properties are usually attributed to *all* members of a premise category and interest is focused on factors that cause shifts in the probability that the property will generalize, rather than on whether probability estimates are normative. Some recent studies (e.g., Lawson & Kalish, 2009), however, have begun to examine feature induction involving more complex

probabilistic relationships between features, premise and conclusion categories, and how inductive predictions are affected by a reasoners' sampling assumptions.

2. It could be argued that many studies of children's induction (e.g., Gelman & Markman, 1986) are exceptions to this generalization since much of this work has been devoted to examining how children generalize the novel properties of specific exemplars rather than categories.